10 **(a)** State Hubble's law. Identify any symbols that you use.

.....

.....

.....

..... [2]

(b) A star of luminosity 3.8×10^{31} W is a distance of 1.8×10^{24} m from the Earth.

Calculate the radiant flux intensity at the Earth of the radiation emitted by the star.

radiant flux intensity = ... $W\,m^{-2}$ [2]

(c) The star in **(b)** is in a distant galaxy. A spectral line in the light from this galaxy is known to have a wavelength of 486 nm. This spectral line in the light from the galaxy observed on the Earth has a wavelength of 492 nm.

(i) Explain why the wavelength observed on the Earth is different from the wavelength that the galaxy is known to have emitted.

.....

.....

..... [2]

(ii) Determine a value for the Hubble constant H_0.

H_0 = ... s^{-1} [3]

[Total: 9]

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